Extra Bite #18: Antenna Matching and Phasing Lines

Gamma / beta / T / stub matches, Q-section impedance, Wilkinson dividers, reflection coefficient, and phasing lines.

Group E9E · 10 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

The Four Classic Yagi-Match Schemes

When a Yagi's driven element does not present 50 Ω at the feed point, you match it with one of these systems. Each looks different on the boom and each has its own quirks.

Match	What it is and notable rule	Q
Gamma	Connect the coax shield to the **center** of the element and the center conductor a fraction of a wavelength to one side. (E9E02) A series capacitor in the gamma rod **cancels unwanted inductive reactance** (E9E04). The element stays bonded to the boom.	E9E02, E9E04
Beta (hairpin)	Shunt inductor across the feed point. **Requires the driven element to be insulated from the boom** (E9E01) and the driven element's feed impedance must be **capacitive — electrically shorter than 1/2 λ** (E9E05).	E9E01, E9E05
T-match	Symmetrical version of the gamma — two tap points, balanced. (Mentioned in the pool as a distractor.)	—
Stub match	**A short length of transmission line connected in parallel with the feed line at or near the feed point** (E9E03). The stub adds a tunable reactance to cancel the antenna's reactance.	E9E03

Q-Section Quarter-Wave Matching

A **Q-section** is a quarter-wave length of transmission line whose characteristic impedance is the geometric mean of the two impedances you want to match:

$Z_Q = \sqrt{(Z_1 \times Z_2)}$
Example (E9E06): match a 100 Ω antenna feed point to a 50 Ω line. $Z_Q = \sqrt{(100 \times 50)} = \sqrt{5000} \approx$ **70.7 Ω**. **75 Ω** coax is the closest standard impedance — the right answer.

Shunt-Feeding A Grounded Tower

Which match shunt-feeds a grounded tower at its base? Gamma match (E9E09). The tower itself is the antenna; the gamma rod taps in just above ground.

Reflection Coefficient

What parameter describes the interaction of a load and a transmission line? Reflection coefficient (E9E07). The ratio of reflected to incident wave at the load; zero for a perfect match, magnitude 1 for total reflection (open or short).

Wilkinson Power Divider

A **Wilkinson divider** is a passive RF splitter that hands the same power to two output ports while keeping each port matched to 50 Ω.

Use for a Wilkinson divider: Divide power equally between two 50-ohm loads while maintaining 50-ohm input impedance (E9E08). Common in stacked antenna systems and in power-amplifier combining.

Phasing Lines

Connecting multiple driven elements with **phasing lines** sets the phase relationship between them at the feed point, which is how an array's directional pattern is created.

Purpose of multiple driven elements connected through phasing lines: To control the antenna's radiation pattern (E9E11). Same principle behind end-fire pairs, broadside arrays, and phased verticals.

Mental Cheat Sheet

- Gamma match: shield to center, center conductor offset; series cap cancels reactance.
- Beta/hairpin: driven element **insulated from boom**; feed Z must be **capacitive**.
- Stub match: short parallel line near the feed point.
- Q-section: quarter wave, Z = $\sqrt{(Z1 \cdot Z2)}$. 100 ↔ 50 → use **75 Ω** line.
- Shunt-feed a grounded tower → **gamma**.
- Reflection coefficient describes load/line interaction.
- Wilkinson splits 50 Ω into two equal 50 Ω outputs.

Common Traps

- The **beta** (hairpin) match — not gamma or T — needs the driven element insulated.
- Gamma's series capacitor cancels **inductive reactance**, not "provides DC isolation."
- The beta match requires a **capacitive** feed point (element shorter than 1/2 λ), not inductive.
- Quarter-wave Q-section impedance is the **geometric mean**, not the average.
- Reflection coefficient — not VSWR alone — is the parameter that "describes load/line interaction."

- Phasing lines on multiple driven elements → control the **radiation pattern**.

- Wilkinson maintains 50 Ω at all three ports; it does **not** match high to low Z.
- Phasing lines shape the **pattern**; they are not for harmonic suppression or band-changing.

Bite 18 Question Bank (E9E)

E9E01. Which matching system for Yagi antennas requires the driven element to be insulated from the boom?
A. Gamma.
B. Beta or hairpin.
C. Shunt-fed.
D. T-match.

E9E02. What antenna matching system matches coaxial cable to an antenna by connecting the shield to the center of the antenna and the conductor a fraction of a wavelength to one side?
A. Gamma match.
B. Delta match.
C. T-match.
D. Stub match.

E9E03. What matching system uses a short length of transmission line connected in parallel with the feed line at or near the feed point?
A. Gamma match.
B. Delta match.
C. T-match.
D. Stub match.

E9E04. What is the purpose of the series capacitor in a gamma match?
A. To provide DC isolation between the feed line and the antenna.
B. To cancel unwanted inductive reactance.
C. To provide a rejection notch that prevents the radiation of harmonics.
D. To transform the antenna impedance to a higher value.

E9E05. What Yagi driven element feed point impedance is required to use a beta or hairpin matching system?
A. Capacitive (driven element electrically shorter than 1/2 wavelength).
B. Inductive (driven element electrically longer than 1/2 wavelength).
C. Purely resistive.
D. Purely reactive.

E9E06. Which of these transmission line impedances would be suitable for constructing a quarter-wave Q-section for matching a 100-ohm feed point impedance to a 50-ohm transmission line?
A. 50 ohms.
B. 62 ohms.
C. 75 ohms.
D. 90 ohms.

E9E07. What parameter describes the interaction of a load and transmission line?
A. Characteristic impedance.
B. Reflection coefficient.
C. Velocity factor.
D. Dielectric constant.

E9E08. What is a use for a Wilkinson divider?
A. To divide the operating frequency of a transmitter signal so it can be used on a lower frequency band.
B. To feed high-impedance antennas from a low-impedance source.
C. To divide power equally between two 50-ohm loads while maintaining 50-ohm input impedance.
D. To divide the frequency of the input to a counter to increase its frequency range.

E9E09. Which of the following is used to shunt feed a grounded tower at its base?
A. Double-bazooka match.
B. Beta or hairpin match.
C. Gamma match.
D. All these choices are correct.

E9E11. What is the purpose of using multiple driven elements connected through phasing lines?
A. To control the antenna's radiation pattern.
B. To prevent harmonic radiation from the transmitter.
C. To allow single-band antennas to operate on other bands.
D. To create a low-angle radiation pattern.

Answer Key

E9E01: B	E9E02: A	E9E03: D	E9E04: B	E9E05: A
E9E06: C	E9E07: B	E9E08: C	E9E09: C	E9E11: A